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Abstract

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A Physically Grounded Discrete Causal Recording Formalism Based on Spatiotemporal Isolation for Surface Propagation Pattern Reconstruction

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Abstract

This study introduces a pattern-oriented mathematical framework, termed the causal recording framework, which is conceptually rooted in cellular automata and consists of a discrete state layer that encodes physical information. Unlike Laplacian-based continuous partial differential equations (PDEs), which are successfully established for domain-dependent macroscopic pattern generation, the causal recording framework generates and records each local state solely from previously registered causal states, regardless of domain size, while incorporating time irreversibility and dissipative structures. The framework is applied to deterministic geometric order on liquid surfaces and validated through direct experiments and pattern reconstruction. Numerical pattern reconstructions demonstrate that the framework can reconstruct the evolution of real water ripple patterns. This research also reports that disordered, symmetry-breaking ripples on water surfaces can evolve into highly organized symmetric water patterns. This transition is reconstructed through a single causal recording framework, suggesting that emergent regularity arises from the interaction of identical local rules without any modification of the governing equations. The results also suggest that spatial confinement and local source information are critical factors in transforming disordered asymmetric patterns into highly ordered Faraday-wave-like patterns. These findings demonstrate the validity of the causal recording formulation as an effective framework for describing natural pattern formation characterized by domain independence, dissipative structures, and time irreversibility.

Introduction

Patterns are ubiquitous in nature as the form of periodic shapes, color, movements, informational rules and rhythms. There patterns appear in a variety of contexts, ranging from living and non-living system, static and dynamic, to causal relationships and spatial order [1-5]. Traditionally, geometric patterns have been modeled using partial differential equations (PDEs) with the Laplacian operator. Alan Turing's reaction-diffusion (RD) model proposed the spontaneous emergence of spatially periodic patterns, providing a framework for understanding biological structures. [6-9] The domain expansion effects on pattern selection and stability have been studied in growing-domain RD systems. [10, 11] The propagation of harmonic surface water ripples, a representative dynamic pattern in the natural world, is well described by the wave equation, the Laplacian, and boundary conditions. [12]

While these equations are mathematically robust and highly effective for domain-dependent macroscopic pattern description in many areas, this Laplacian-based continuous PDE formulation exhibits innate limitations for real-time reconstruction of natural patterns. It is governed by global coupling that induces globally interacting continuous fields and domain-dependent property alterations. In addition, maintaining these continuous fields across expanding domains or higher resolutions increase a substantial computational cost. Consequently, these limitations hinder the natural representation of patterns governed by localized causal histories, ranging from the irreversible, history-dependent pattern formation observed in solid crystal growth to the asynchronous interference of multi-source water ripples generated by discrete raindrop impacts.

Moreover, these computational and structural bottlenecks have limited the direct, continuous comparisons between the simulated propagation process and actual pattern formation.

The macroscopic nature is inherently governed by dissipative structures—open systems that operate through continuous processes across localized regions. In such environments, energy is not merely a conserved quantity but inevitably diminishes when interacting with the medium during propagation. [13-15]. Real-world examples of this causal progression are abundant, from the conversion of thermal energy into mechanical work in a steam engine to the sudden release and propagation of kinetic energy during a seismic event, the precipitation of a metal complex from a salt solution, and the intricate transformation of mechanical stress into chemical potential in mechanochemical reactions [16-20]. This process is not merely a continuous evolution but one with a beginning and an eventual rest as the system dissipates. Every flow of force and energy during pattern generation in the macroscopic world is an inherently time-irreversible, sequential phenomenon generated by a spatio-temporally isolated local origin where the initial force is injected and then propagated. There exists a definitive origin of force generation: a specific moment and location where force is first initiated and propagated, and where it disappears as it transitions.

Cellular automata (CA) model consist of many simple identical components is capable of generating remarkably complex patterns [21]. The fundamental strength of the CA model lies in local interactions of local interactions according to predefined rules. Furthermore, the versatility of CA as a tool for dynamic modeling has been investigated across a wide range of fields for simulating complex systems in biology, computer science, and particularly materials science [22]. This local-to-global transition aligns with my observation that mechanical events are propagated and recorded at a finite velocity through adjacent points in a medium.

In addition, the Lorenz system is widely recognized as a foundational model for explaining chaos and complex nonlinear dynamics [23]. While it is typically studied for its sensitivity to initial conditions, I focus on two specific attributes of this system. First is the inherent causality, which establishes a rigorous link between temporal states [24]. Second, I focus on the system's ability for autonomous evolution governed by internal rules and initial state of local source. In this view, complex patterns can be emerged based solely on its own deterministic equations, without the need for continuous external intervention. Despite their success as theoretical frameworks, these two models face inherent challenges in directly modeling or describing actual natural phenomena. their abstraction often limits the integration of real-world physical constants and variables, such as mass, velocity, etc. which are essential for a comprehensive representation of physical world.

This work introduces a discrete state layer responsible for propagating locally defined physical information through causal recording with state transitions governed by vector-based rules grounded in Newtonian dynamics and Hooke's law, enabling the reconstruction of experimentally observed water-surface patterns. The model is constructed from four fundamental principles: (1) localized injection of physical force at isolated origins, (2) operator-based transformation of physical information into the discrete state layer, (3) causal force propagation and geometric recording of force history across adjacent domains, and (4) local regeneration at interaction boundaries. Through this approach, this study investigate whether localized information propagation—inherently endowed with time irreversibility, a definitive life cycle, and dissipation—can reconstruct experimentally observed water-surface ripples while preserving essential physical information. This work also demonstrate that the proposed formulation gives rise to the spontaneous emergence of symmetric surface patterns with spatial confinement, continuous force input, and multi-source interactions.

Results

Conceptual origin of deterministic Water Surface Pattern Formation

In this work, the present study sees to find the fundamental cause of water surface pattern formation in the spatio-temporal recording of local inputs, motivated by the 'Walking droplets' experiment. Successive collisions of liquid droplets generate repetitive ripples on the surface, and

the droplets move forward by interacting with the wave structures by their past inputs [25]. It can be interpreted that continuous structures emerge from an initial stimulus as a historical recording.

In this system, the observed water surface response is decomposed into four sequential mechanisms: the generation of a local source force, operator-based discrete layer mapping, causal force propagation with geometric recording of force history, and the regeneration of the source triggered by the obstruction of front propagation at spatial boundaries (Fig. 1). The collective surface behavior emerges from the time-ordered execution of these rule-based local mechanisms. Consequently, the apparent propagation of water surface patterns during propagation reflects a recorded sequence of local force states.

Local source force refers to a localized mechanical input applied to the water surface, acting on a discrete area to establish a clear force direction. Motivated by the periodic impacts in 'walking droplets' experiments, this initial stimulus triggers a deterministic response within the water surface. Upon application, the surface element undergoes a mechanical displacement, and a restoring response arises due to elastic and gravitational effects inherent to the fluid interface. This interaction sets the fundamental scale and orientation for subsequent pattern formation. The response of the local site is governed by predefined mechanical rules, such as inherent stiffness and mass, which effectively encode the input into a historical state. In this context, the source force serves as the geometric and causal origin providing the temporal recording of the complex water surface patterns emerge through autonomous evolution.

Operator-based discrete layer mapping determines the generated local source force is registered into the discrete layer during propagation. Once a force state is recorded, the operator establishes an irreversible barrier that prevents backward transmission, thereby preserving spatio-temporal causality and directional front advancement.

Force transfer and geometric recording of force history describe how the mechanical influence of a localized source force manifests at neighboring surface regions over time. Following the initial input, adjacent areas of the water surface exhibit delayed mechanical responses governed by a finite propagation speed. This phenomenon defines the system's causal horizon. Importantly, this process does not involve the continuous transport of a physical entity; it reflects a sequential recording of local force states. Directionally encoded force vector is systematically transformed into kinetic responses at each site as the causal front passes. Each water surface element begins its own autonomous evolution by the identical deterministic rules as the source, ensuring the causal continuity of the emergent water surface pattern without sustained external intervention.

Force regeneration indicates the re-emergence of surface motion following mechanical interaction with spatial boundaries or geometric constraints. It denotes the continued execution of the original rule-based sequence under restricted conditions. When the propagating front reaches the edge of this defined functional domain, the spatial propagation is obstructed, and the local state is reconfigured. At this interface, the incoming kinetic response is deterministically converted into a downward directed force. This stored state then serves as a localized causal trigger, allowing the water surface pattern to persist and evolve autonomously.

A local deterministic causal framework of surface records

The mathematical framework proposed here builds upon the concept of sequential recording of the local state introduced in our previous study of a temporal recording formulation for surface pattern propagation [26], while extending it into a causal recording formulation for deterministic pattern formation. In that work, an eikonal-type formulation was used as a temporal registration device, encoding when a given spatial location becomes mechanically eligible to respond to a localized force input [27].The present study builds directly on that conceptual foundation, while extending it to a fully deterministic force–mass framework applicable to real water surfaces. The conceptual roots of the present formalism lie in cellular automata, particularly in their use of local update rules and sequential state generation. Unlike conventional cellular automata, however, the proposed framework introduces physically parameterized recordings and an event-driven switch operator $H(\nabla\psi)$, enabling reconstruction of experimentally observed pattern structures. The eikonal front is retained not as a wave construct, but as a causal horizon that governs the temporal ordering

of local mechanical responses. This allows the surface evolution to be described through a sequence of local force applications, transfers, and regeneration governed by Newtonian dynamics and Hooke's law.

The full governing equation introduced below is constructed to reflect the three mechanisms outlined in the system description: (i) localized source force generation, (ii) operator-based discrete layer mapping, (iii) temporally ordered force transfer and geometric recording of force history, defined by a finite propagation rule, and (iv) deterministic regeneration through force conversion within the local front mass. The governing equation is formulated to encode how a localized mechanical input becomes accessible across space through time and is locally executed by the surface element.

$$\Psi(r,t) = A\alpha(\tau)H(\phi)$$

In this equation, $\Psi(r,t)$ represents the recorded mechanical state of the water surface at position r and time t. Importantly, ψ does not denote a transported physical quantity, but the outcome of a locally executed force–mass interaction conditioned by the accessibility of prior force history.

Source Force Term A

The coefficient A represents the magnitude and spatial distribution of a localized mechanical input applied to the water surface at position r_s. In this framework, the source force is defined by the interaction between the input magnitude and a structural domain, expressed as:

$$A(r) = A \times \beta(r)$$

The introduction of the spatial β field defines the external state of the space through which the force propagates. The β field modulates how local environmental factors condition the progression of the source force. In this study, the β field is implemented as a binary domain (β = 1 or 0) to investigate the system's response under the most extreme environmental conditions. When the initial force is applied, it acts on a local surface element and establishes a well-defined initial direction for the surface displacement. This interaction sets the characteristic scale and geometric orientation for all subsequent motion. The evolution of the pattern emerges entirely from the deterministic mechanical response of each discrete site within the permitted spatial domain.

Local Mechanical Response $\alpha(\tau)$

The function $\alpha(\tau)$ describes the time evolution of the local surface response following force application:

$$\alpha(\tau) = e^{-\gamma\tau}\cos(\omega\tau), \quad \text{with } m\cdot\ddot{a} + c\cdot\dot{a} + k\cdot a = 0$$

This term corresponds to the solution of a damped harmonic response governed by the effective mass m, damping coefficient c, and restoring stiffness k of the water surface. Physically, it represents how a displaced surface element undergoes restoring motion due to gravitational and elastic effects. The response is entirely local and deterministic, and its temporal evolution is recorded rather than propagated.

Causal Horizon Operator $H(\phi)$

The term $H(\phi)$ defines the spatiotemporal accessibility of the recorded force history within the discrete state layer, acting as a causal operator that determines which previously recorded physical information can be accessed by subsequent local states.

$$H(\phi) = H\left(t - \frac{|r - rs|}{V}\right)$$

Where V is defined as an intrinsic speed constant for the transmission of physical force through space, governing how rapidly that specific causal signal can geometrically register its presence across the domain. This operator determines whether a given surface location can access the force history up to time τ. As a result, different spatial locations execute identical mechanical rules over different accessible time intervals.

$$t(r) = \int_0^T dt_{local}(r), \qquad dt_{local}(r) = dt_0 \times \beta(r)$$

Instead of treating time as a rigid, the local time step dt_{local} and resulting physical time $t(r)$ are dynamically scaled by the local β field.

Within this framework, the apparent propagation of water surface patterns arises from the spatial expansion of temporally accessible force histories. Each surface element responds by executing the same local force–mass relation once the causal condition is satisfied. When the accessible force history reaches a geometrical boundary, the boundary initiates a new localized source force, restarting the same deterministic process under altered geometric constraints. This regeneration mechanism preserves the directionality of the force–mass interaction while allowing continued surface propagation.

The integration of the Heaviside function $H(\phi)$acts as a causal switch within the discrete state layer, ensuring an orderly spatio-temporal progression via a constant characteristic velocity V. This invariant speed allows the system to preserve the original force history across the surface without numerical distortion, providing as a consistent causal recording process.

Pattern reconstruction of water surface ripples by causal recording formulation

Fig. S1(A) and movie S1 present the pattern reconstruction of single surface evolution generated solely by the governing equation of local element, without imposing any boundary-induced regeneration or external constraints. A single localized source force is applied at the center of the domain, and the subsequent surface response is computed according to the deterministic and causal accessibility defined in the model. The upper panels show the two-dimensional surface record at increasing times. As time progresses, a radially expanding surface pattern emerges from the source location. This expansion arises from the sequential activation of local surface elements as their accessible force history becomes available. The spatial extent of the pattern increases monotonically with time, that reflect the finite speed at which causal accessibility expands across the surface. The lower panels display one-dimensional cross-sectional profiles taken through the center of the domain. These profiles reveal a structured alternation in the recorded surface state as a function of position. In addition, each surface element executes a time-dependent mechanical

response once activated. Notably, the spatial repetition observed in the cross-sections is not preassigned but emerges from the combination of the local restoring force vector and the temporal ordering imposed by the predefined rules of the initial source. A characteristic feature of the pattern reconstruction for ripple is the asymmetric structure at the leading edge of the expanding surface record. The front exhibits a truncated or partially developed structure. This behavior reflects that surface elements near the advancing front access only a limited portion of the force history and simultaneously experience a spatially distributed reactive force arising from local mechanical recoil. As a result, the local response near the front remains geometrically and temporally incomplete compared to regions closer to the source.

Fig. S1(B) and movie S2 examine how variations in the initial mechanical conditions influence the resulting surface pattern record. Compared to Fig. S1(A), the input force is increased, the effective mass is reduced, and finite damping is introduced. These changes localize the initially perturbed surface region while enhancing the directional magnitude of the response. Under these conditions, the leading surface record exhibits a pronounced directional amplitude aligned with the initial force orientation, whereas subsequent surface responses are strongly decreased. This contrast indicates that the earliest activated surface elements retain a dominant portion of the force history, while later responses are suppressed due to geometric dispersion and damping. The cross-sectional profiles emphasize that the spatial distribution of recorded surface states is highly sensitive to the initial force–mass configuration. Small changes in the applied force and local inertia can produce qualitatively different surface evolution patterns, even under identical governing rules. This result highlights that surface propagation in the present framework depends on the mechanical conditions under which the source force is applied.

Fig. 2(A) and movie S3 show the surface record produced by a single localized source force under finite damping. Fig. 2(B) and movie S4 display the surface records generated by two localized source forces, identical compared to the source of Fig. 2(A). The resulting patterns exhibit spatial regions where local mechanical responses are enhanced, as well as regions where they are locally diminished. These features arise without the introduction of additional mathematical rules or coordination mechanisms and follow directly from the interaction of independently executed local force histories. It is shown that compatible directional tendencies of accessible force records reinforced the local response. In contrast, the coexistence of opposing directional tendencies reduced the local force vector. This behavior indicates that the mechanical record retains directional information and that local force executions can coexist and interact in space. Consequently, the resulting pattern exhibits structural features analogous to wave interference in fluid dynamics. This emergent behavior demonstrates that complex superposition-like effects can be reproduced through the local interaction of independent force histories, confirming that the deterministic causal framework inherently accounts for the reinforcing and canceling behaviors characteristic of water surface interference.

Fig. S2 and movie S5 validates the functional role of the β field as a spatial partitioning operator that restricts causal propagation. The governing rules are focusing on whether the β field successfully terminates the expansion of the surface record at predefined limits. The side-view profiles clearly demonstrate that the causal front is abruptly arrested upon encountering the β-boundary (0). This result confirms that the structural domain effectively determines the spatial allowness of local surface elements.

Fig. 3, movie S6 and movie S7A compare ripple-like surface records generated by the present formulation introducing regeneration force with experimentally observed water surface behavior under comparable geometric conditions. The regeneration mechanism was implemented via a coding algorithm designed to trigger a new source force at the moment the causal front makes contact with the spatial boundary where the β-field is defined as zero or where propagation is arrested. The regenerated force was assigned a magnitude equivalent to 50% of the input source force to account for energy dissipation during boundary interaction.

In panels (B) and (D), water droplets are introduced at the side and center of a confined container, respectively, and the resulting surface motion is monitored in real time. Panels (A), (C) show the

corresponding numerical ripple-like surface records produced by a single localized source in geometrically similar domains, without introducing additional boundary rules. The simulated ripple surface exhibit spatial and temporal patterns that closely resemble those observed in the experiments, constrained by the defined β field. The geometric organization of surface structures imposed by the container boundaries is reflected in both the experimental images and the recorded force patterns generated by the model. This correspondence indicates that the present formulation captures key mechanical features governing water surface behavior through local force execution and temporal ordering alone. The observed correspondence suggests that describing water surface motion as a time-recorded execution of local mechanical interactions provides a natural and effective framework for interpreting experimentally observed surface patterns. Across both experiment and pattern reconstruction, the governing formulation is unchanged. The only distinction lies in the spatial placement of the initial localized event, from which all subsequent surface patterns emerge through time. I also analyzed the structural consistency of the emergent water ripple pattern under varying local time steps ($dt = 1.0, 1.5, 0.7$), while all other parameters are kept identical (Fig. S3, movie S7B ~ S7D). The geometric configuration and complex interaction of the generated water patterns by the causal recording framework exhibit invariance across all three regimes. The only variation is the physical step count required to achieve the corresponding evolutionary state in exact proportion to the dt variation. This consistency confirms that the framework treats dt as a pure scaling operator of the temporal progression of water propagation. Overall, the causal recording formulation successfully reconstructs surface patterns that closely match real-time water ripples. Furthermore, modulating the initial spatial positions and temporal intervals of independent sources confirms that diverse surface patterns can be generated, establishing a robust foundation for examining more complex collective behaviors.

Emergence of Faraday-wave-like symmetric patterns under restricted spatial boundaries

Building upon the model's success in reproducing real water surface behavior, surface evolution under highly restricted spatial accessibility is explored. The causal recording formulation maintains spatio-temporal isolation between advancing states, renders global interactions between individual sources nonexistent, and is distinguished from boundary-scale distortion. Thus, it becomes feasible to investigate complex pattern formation under confined boundaries subjected to continuous, randomized multi-source inputs. In this context, the emergence of regular surface structures, resembling Faraday-like waves, is driven by two key factors: stringent spatial confinement and the sustained presence of force inputs. To simulate the continuous vertical oscillation characteristic of Faraday waves, the damping coefficient in this numerical framework was fixed at zero.

Fig. 4 present the surface records generated by the present formulation when the spatial domain is geometrically confined. The two-dimensional surface records exhibit repeating geometric patterns, while the corresponding one-dimensional cross-sections reveal regular alternation in the recorded mechanical state (Fig. 4(A) and movie S8). While local-rule-based patterns often exhibit symmetry-breaking behavior in open domains, the introduction of geometric constraints forces the accumulation and interaction of directionally encoded force histories into a highly organized state. As a result, the recorded mechanical states transition from irregular fluctuations to well-defined, spatiotemporally symmetrical patterns. Notably, this transition is achieved within the same governing framework, without any modification of the underlying equations or the addition of external nonlinear terms.

Fig. 4(B) and movie S9 demonstrate that the resulting surface patterns are highly sensitive to the initial mechanical rules. A subtle shift from 0.5 to 0.3 in stiffness (k) qualitatively alters the recorded geometric configuration, highlighting that the emergent order is strictly governed by the local deterministic parameters rather than being a mere reflection of the boundary shape. As the spatial domain expands in Fig. 4(C), (D), and movie S10, S11, the surface records evolve into increasingly complex yet strictly symmetrical geometric patterns. Despite this growing global complexity, the central core consistently preserves the fundamental pattern observed in Fig. 4(A)

throughout the temporal evolution. This persistence suggests that the local mechanical record maintains its causal integrity at the origin, even as the extended domain allows for a denser accumulation and interaction of force histories. In Fig. 4(E), (F), and movie S12, S13 the system is simulated under a distributed force environment where sources are uniformly applied across the domain.

Notably, Fig. 4(E) shows that the resulting symmetrical pattern remains nearly identical to Fig. 4(D) regardless of the number or arrangement of sources, suggesting that the initial physical state and spatial constraints are more fundamental in shaping the surface behavior than the specific positioning of inputs. Furthermore, Fig. 4(F) demonstrates that minimal parameters (the local mass, element density) change leads to the emergence of highly symmetrical crystalline-like patterns. This phenomena confirms that the deterministic force-mass framework can effectively capture such sophisticated physical transitions. The observation of symmetric pattern transitions through variations in m and k can be interpreted as a phenomenon analogous to the instabilities observed in physical experiments, where patterns evolve according to the liquid medium (density/viscosity) and the vertical driving frequency. Interestingly, the patterns observed in Fig. 4(D) at t=750 and t=1250, as well as in Fig. 4(F), exhibit a striking resemblance to the experimental Faraday wave patterns observed in a square vessel brimful of water, with the vertical reflection of a light beam, reported in previous literature [28]. Reported results are controlled by local constraints and parametric excitation, the structural transitions in this study can be interpreted as a result of introducing velocity heterogeneity into the propagation front. This suggests that the present pattern reconstruction framework not only replicates the visual outcomes but also effectively encapsulates the underlying experimental conditions where physical parameters dictate the emergent symmetry.

Fig. S4 and movie S14 ~ S16 investigate the influence of stochastic inputs on the emergent surface record by applying random spatial positioning and temporal onset to the sources. While these randomizations lead to a breakdown of spatial symmetry, the system remarkably maintains its temporal symmetry, with each local element continuing to execute its periodic mechanical response. This result confirms that while global spatial order is governed by the boundary-induced confinement and the fundamental temporal rhythm is an intrinsic property of the deterministic local rule. This results highlights that the spatial restriction of the domain remains the dominant factor in ensuring the long-term structural organization of the surface record.

The collective results of Fig. 4 illustrate a distinctive transition from disordered surface fluctuations to highly ordered geometric structures. While the local-rule-based surface pattern reconstruction for ripple exhibits symmetry-breaking behavior, the introduction of spatial confinement forces these independent mechanical histories to interact and synchronize, forming symmetrical patterns. This observed transition suggests that the emergence of global symmetry is a deterministic outcome of the balance between spatial propagation limits and the superimposed interaction of regenerated forces. Consequently, this framework suggests the possibility that asymmetric ripples and symmetrical geometric order can both emerge from the same underlying causal recording rule. Within this perspective, the transition between these surface patterns is not a change in fundamental physics, but a deterministic outcome of how spatial constraints and source information modulate the manifestation of local interactions.

Discussion

The continuous attenuation of both the propagation force and velocity due to external dissipation is a natural phenomenon in the real world. To address this, I introduce an external dissipation term at a conceptual level to improve the scalability of this framework. This mechanism mathematically couples the spatial coordinates with the propagation force (A) and the temporal step (dt). Through numerical simulations, I also verify the continuous operational consistency of this dissipation term and analyze its impact on pattern formation.

The dissipation introduced into the governing equation is expressed as:

$$A(r) = A_0 \times \beta(r) \times \left(\frac{1}{1 + \lambda_1 \times r} \right)$$

λ_1 denotes the external force dissipation coefficient, which mathematically scales the attenuation. This formulation represents a spatial (distance-based) attenuation.

$$dt_{local}(r, \tau) = dt_0 \times \beta(r) \times \left(\frac{1}{1 + \lambda_2 \times \tau} \right)$$

Similarly, λ_2 acts as the temporal dissipation coefficient. In contrast to the spatial decay of the propagation force, this formulation introduces a spatiotemporal-dependent attenuation for the local time step. While the magnitude of dt_{local} is constrained by the spatial distribution of the $\beta(r)$ field, its propagation velocity is dynamically dependent by the temporal variable τ. To verify the validity of the proposed dissipation term, I performed numerical simulation in Fig. S5 and movie S17A ~ 17C. While the emergent water patterns remains consistent across all configurations, their cross-sectional profiles reveal distinct physical dynamics. In Fig. S5(B), the introduction of λ_1 successfully induces a continuous attenuation of the amplitude of the front as the propagation distance increases. Regime (C) demonstrates that λ_2 preserves the force of the front but explicitly introduces a temporal delay in the registration progression. This is proven by the structural equivalence between the state at t = 400 in the baseline model (A) and the state at t = 600 in the model (C). These findings confirm that even when the propagation force and velocity are progressively diminished by external dissipation, the underlying mechanism of pattern generation and causal registration within this framework operates with fidelity.

Conclusion

Using the proposed causal recording formulation, conceptually rooted in cellular automata and built upon a discrete state layer in which physical information is causally recorded through sequential local state generation, this study demonstrates the reconstruction of experimentally observed water-surface pattern generation. The results show that dynamic water-surface patterns emerge deterministically through local causal recording, indicating that complex pattern organization can arise from a single history-driven generative rule. By introducing physically grounded local rules into the eikonal-type causal horizon that operate internally and occur externally within a discrete element, it demonstrates that the causal recording framework can successfully reconstruct real water surface structure evolution.

Furthermore, the formulation guarantees consistent state reproduction under identical initial forces regardless of domain size, with time irreversibility and dissipative structures. In addition, multi-source systems and temporally staggered events can be flexibly introduced simply by appending discrete source lists and modulating their local parameters, without increasing global computational coupling. With these characteristics, this approach enabled the direct investigation of structural pattern formation arising under spatial confinement, continuous force inputs, and multi-source interactions. Unexpectedly, within this causal recording environment, Faraday-wave-like symmetric patterns spontaneously emerged, exhibiting highly ordered surface structures closely resembling experimentally reported Faraday waves. This approach allows for a unified "grammar" that reproduces both simple ripples and complex, Faraday-wave-like symmetric patterns by controlled space area without the need for phenomenological corrections. A central finding of this work is that

spatial confinement acts as a catalyst for order; it transforms the inherently nonlinear and symmetry-breaking tendencies of local rules into spatiotemporally symmetrical patterns.

Based on these results, I anticipate that the proposed framework can investigate the generation of various patterns in nature with localized physical phenomena at scales across varying domain sizes and multi-source interactions, free from artificial boundary coupling, while preserving nature-like time irreversibility and dissipative characteristics.

Materials and Methods

To validate the numerical pattern reconstruction of water ripple patterns, macro-scale fluid experiments were conducted using two types of vessels with distinct boundary surface properties. Experimental water surface patterns were generated inside a hydrophobic polypropylene plastic container (19 × 12 cm) filled with water. Initial surface patterns were introduced by dropping water droplets at localized coordinates precisely corresponding to the source positions defined in the numerical models. The spatiotemporal evolution of the resulting ripples and boundary-driven reflections was captured in real time using a smartphone camera (iPhone 14). Detailed descriptions of the algorithm implementation and comparative movie analyses are provided in the Supporting Information.

Data and Code Availability

The source code and all supplemental movies (Movie S1 to S16) are available at: https://github.com/junhyunglee73/real-space-water-surface-dynamics.git

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Figures and Tables

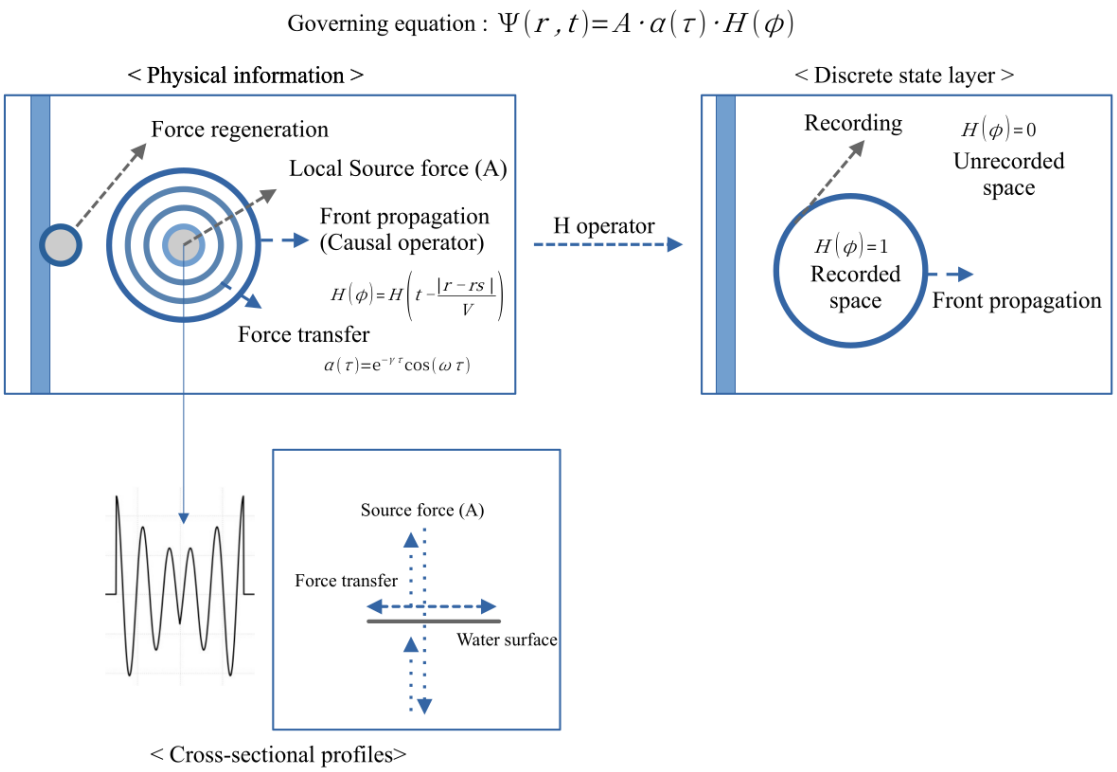


Fig. 1. Conceptual schematic of deterministic water surface pattern formation by causal recordings. The surface response is decomposed into four sequential mechanisms: (i) Injection of an isolated localized source force, (ii) operator-based layer mapping, (iii) causal force propagation and geometric recording of force history, and (iv) local regeneration at boundaries

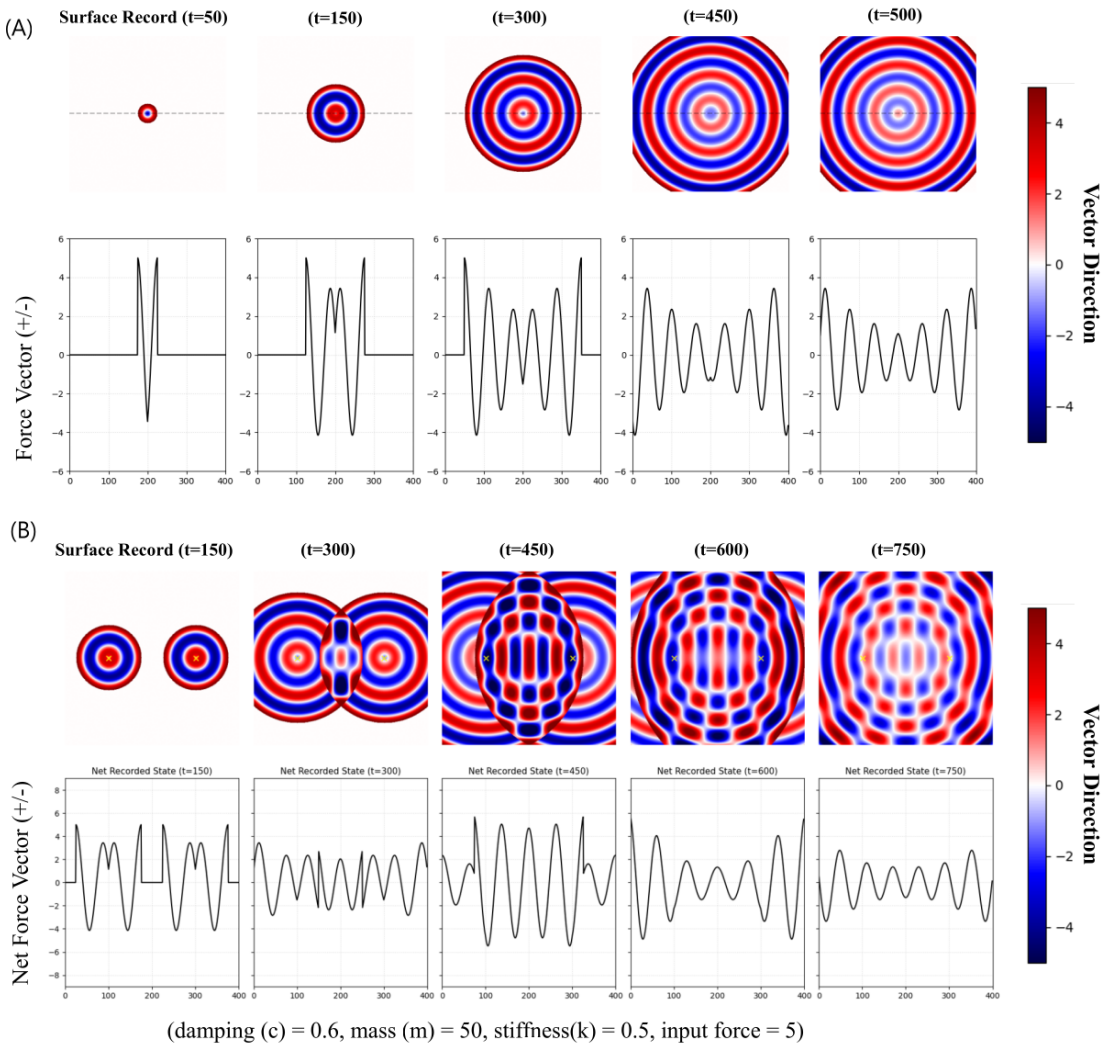


Fig. 2. Emergent interference patterns from interacting causal sources. (A) Surface record of a single damped source. (B) Interaction of two identical sources where compatible force histories reinforce the local response, while opposing histories lead to local cancellation. The resulting pattern demonstrates that wave-like interference emerges deterministically from the spatial coexistence of independent local force executions.

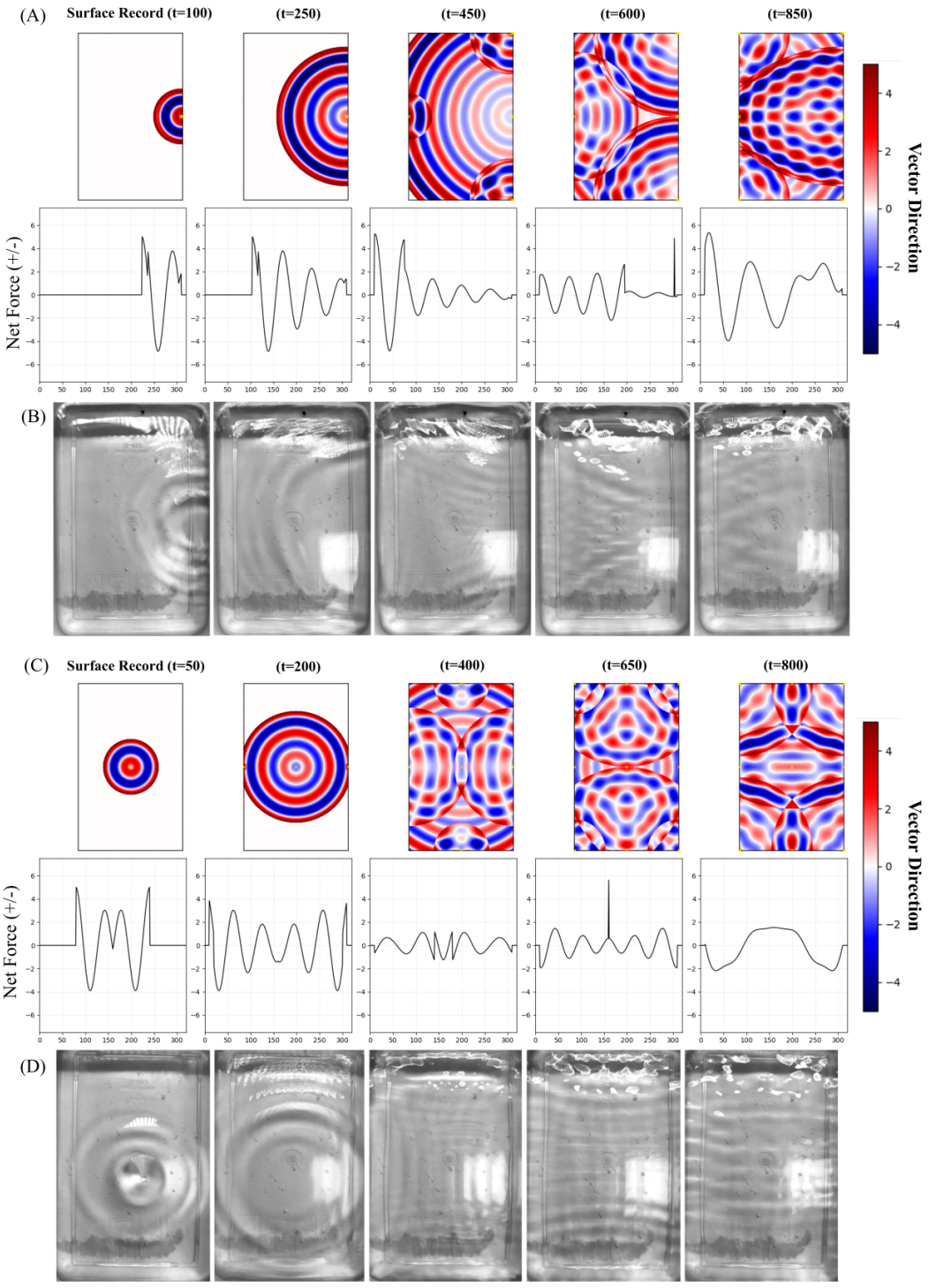


Fig. 3. Comparison between simulated surface records and experimental water surface pattern(A, C) Numerical surface records generated by a single localized source within a restricted -field. (B,D) Corresponding experimental water surface patterns obtained by introducing droplets at analogous positions in a container of similar geometric proportions.

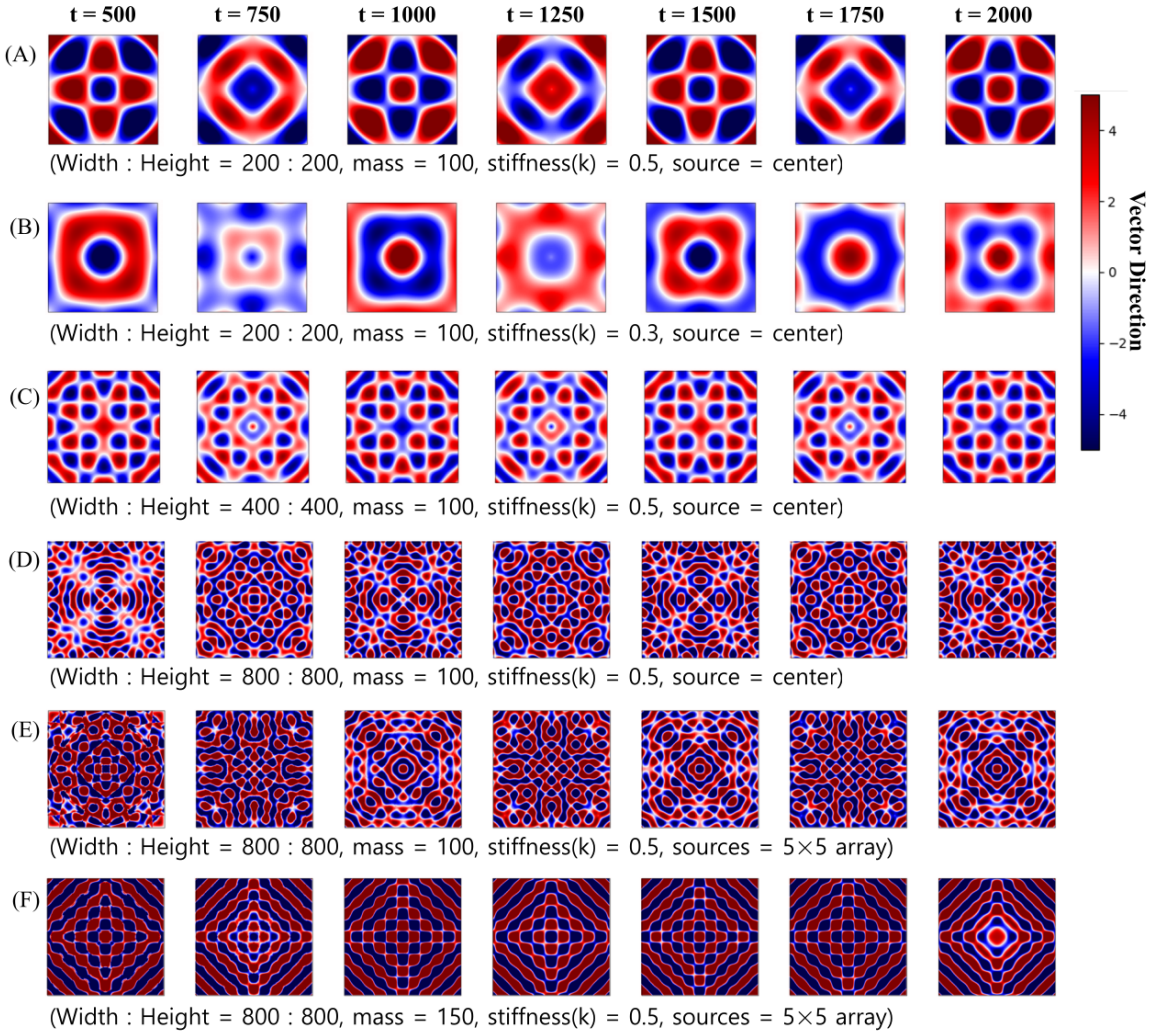


Fig. 4. Emergence of symmetrical geometric order through constrained causal interactions. (A–B) Surface records where subtle shifts in local stiffness () qualitatively reconfigure the pattern density within a fixed -field. (C–D) Expansion of the spatial domain showing increased global complexity while the central core preserves its fundamental structural integrity. (E–F) Collective responses under distributed inputs; (E) demonstrates that spatial constraints dominate the pattern over source positioning, while (F) shows mass and elements density driven transitions to crystalline similar structures. The resulting symmetry represents a deterministic balance between spatial propagation limits and the superimposed interaction of regenerated forces.

Supporting Information for

A Physically Grounded Discrete Causal Recording Formalism Based on Spatiotemporal Isolation for Surface Propagation Pattern Reconstruction

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Experiment

Water surface patterns formation

Water was poured into a hydrophobic plastic (polypropylene) container (19 × 12 cm). The surface patterns were generated by introducing droplets at locations corresponding to those used in the pattern reconstructions. The resulting surface motion was recorded using the camera of an iPhone 14.

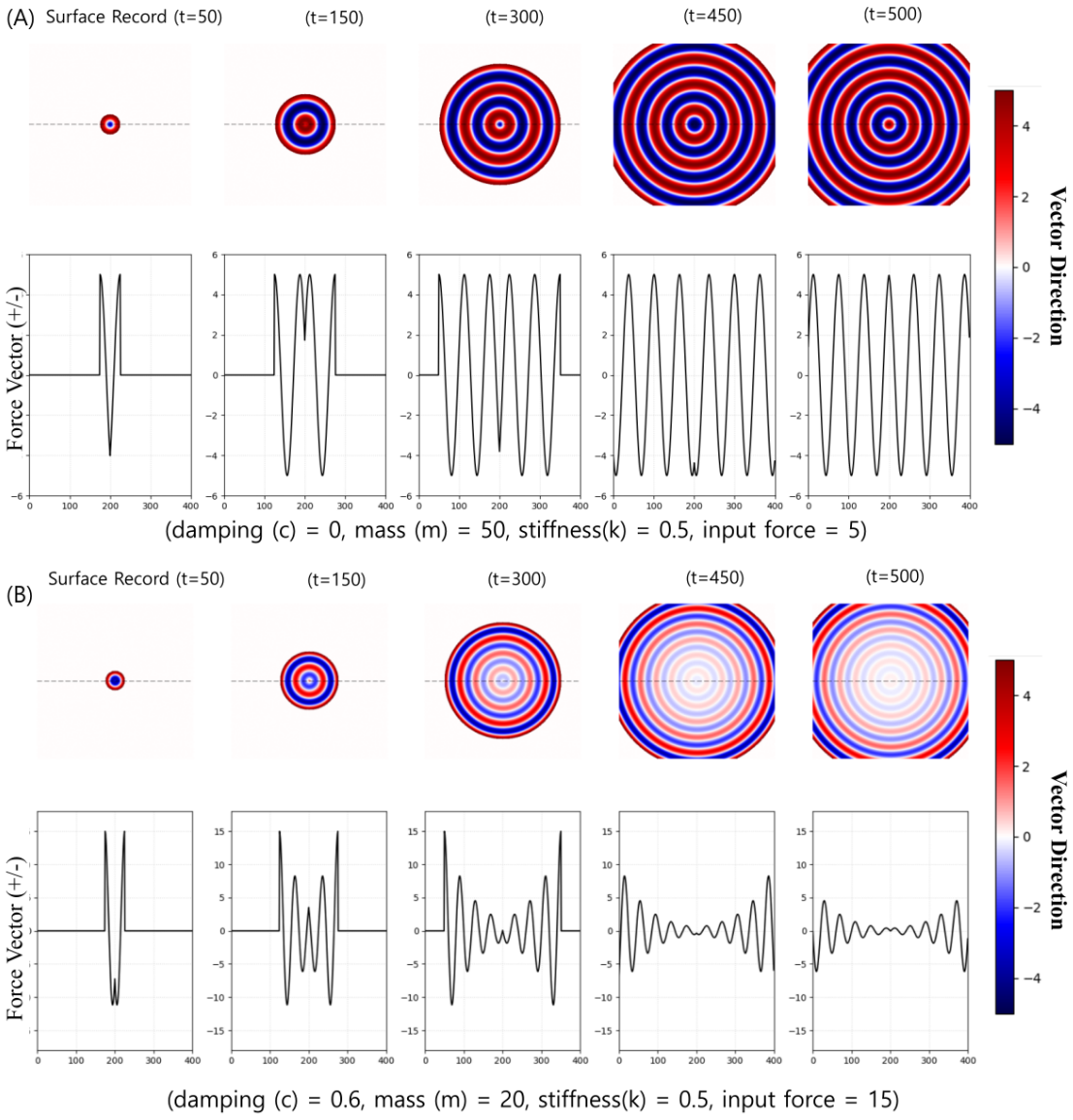


Fig. S1. Emergent surface patterns generated by a single causal source in the absence of boundary regeneration. (A) Time-resolved snapshots of the recorded surface state produced by a localized source force governed solely by the causal record kernel. No boundary-induced source regeneration or additional spatial operators are applied. B) Increasing the source force and reducing the effective mass modifies the directional strength of the initial surface response, demonstrating that the propagation pattern is directly shaped by the input force parameters rather than imposed spatial constraints.

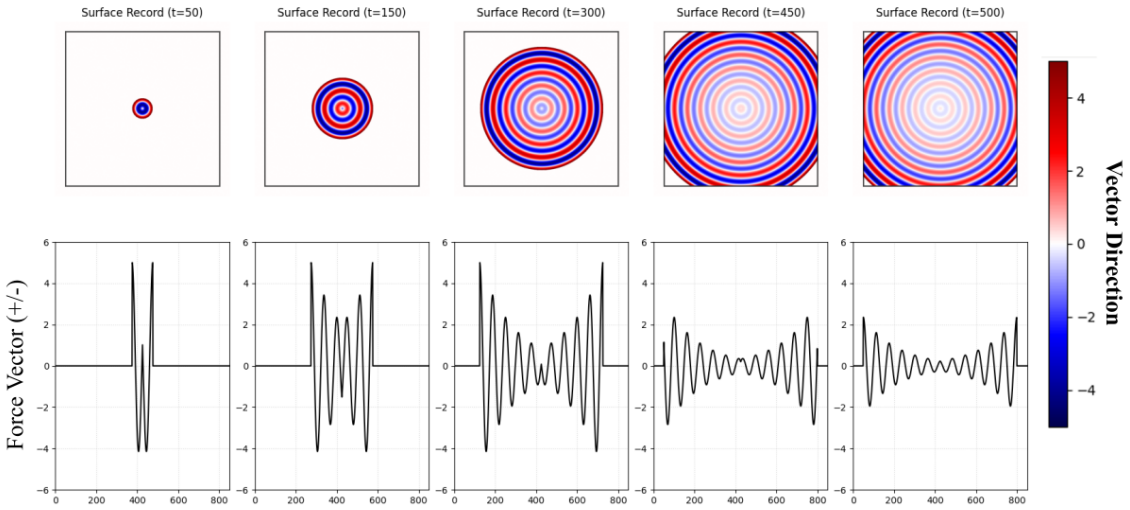


Fig. S2. Validation of spatial propagation arrest via the field. (Top) Sequential snapshots showing the radial expansion of the surface record being terminated at the spatial boundaries defined by the field. (Bottom) Corresponding one-dimensional profiles illustrating the abrupt cessation of the causal front at the boundary.

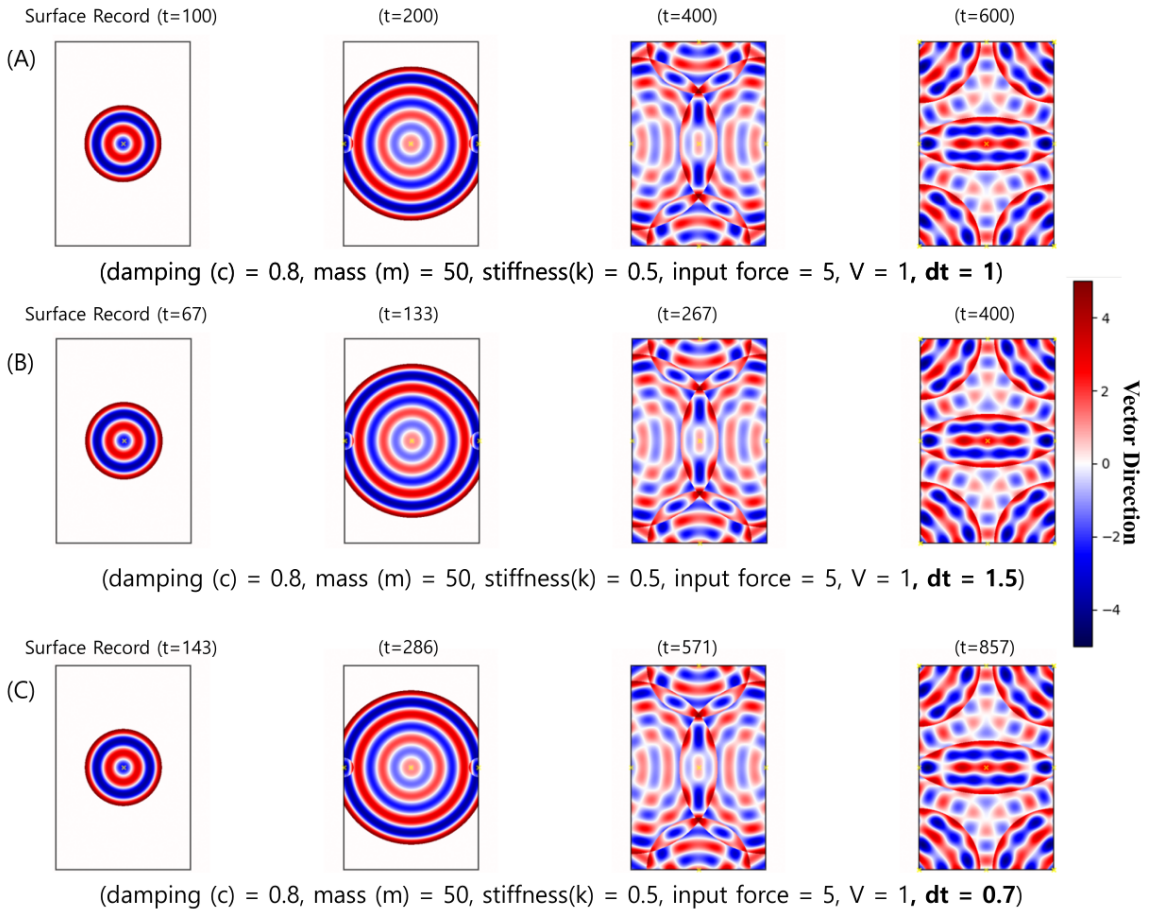


Fig. S3. The emergent surface patterns under varying local time steps (dt). The geometric configuration of the generated patterns remains invariant, demonstrating that dt acts solely as a temporal scaling factor without distorting the final causal recording patterns.

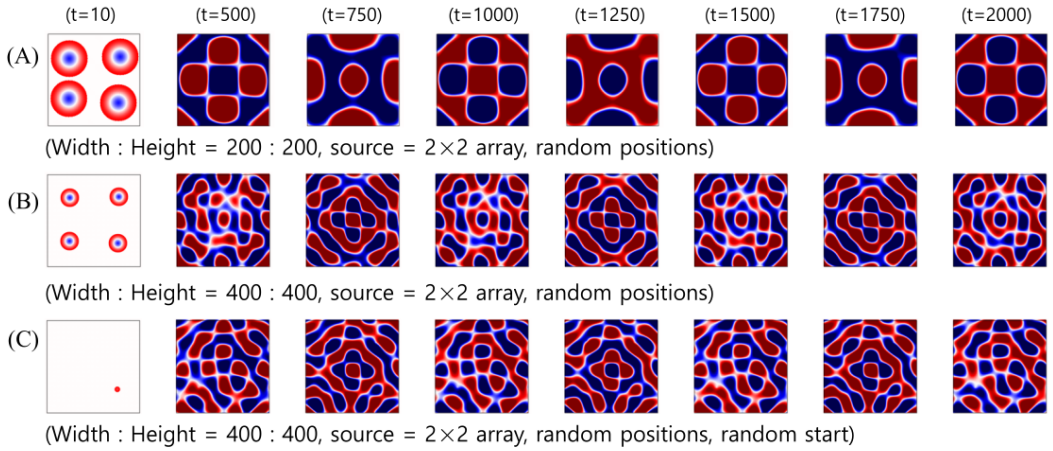


Fig. S4. Robustness of temporal symmetry under stochastic spatial and temporal inputs. Sequential snapshots and profiles of surface records generated by sources with randomized spatial positions and onset times. Although the stochastic nature of the inputs leads to a breakdown of global spatial symmetry, the underlying temporal symmetry is maintained through the independent execution of local mechanical rules. This decoupling confirms that while the spatial domain dominates the geometric organization of the pattern, the characteristic temporal rhythm remains an intrinsic property of the local deterministic framework.

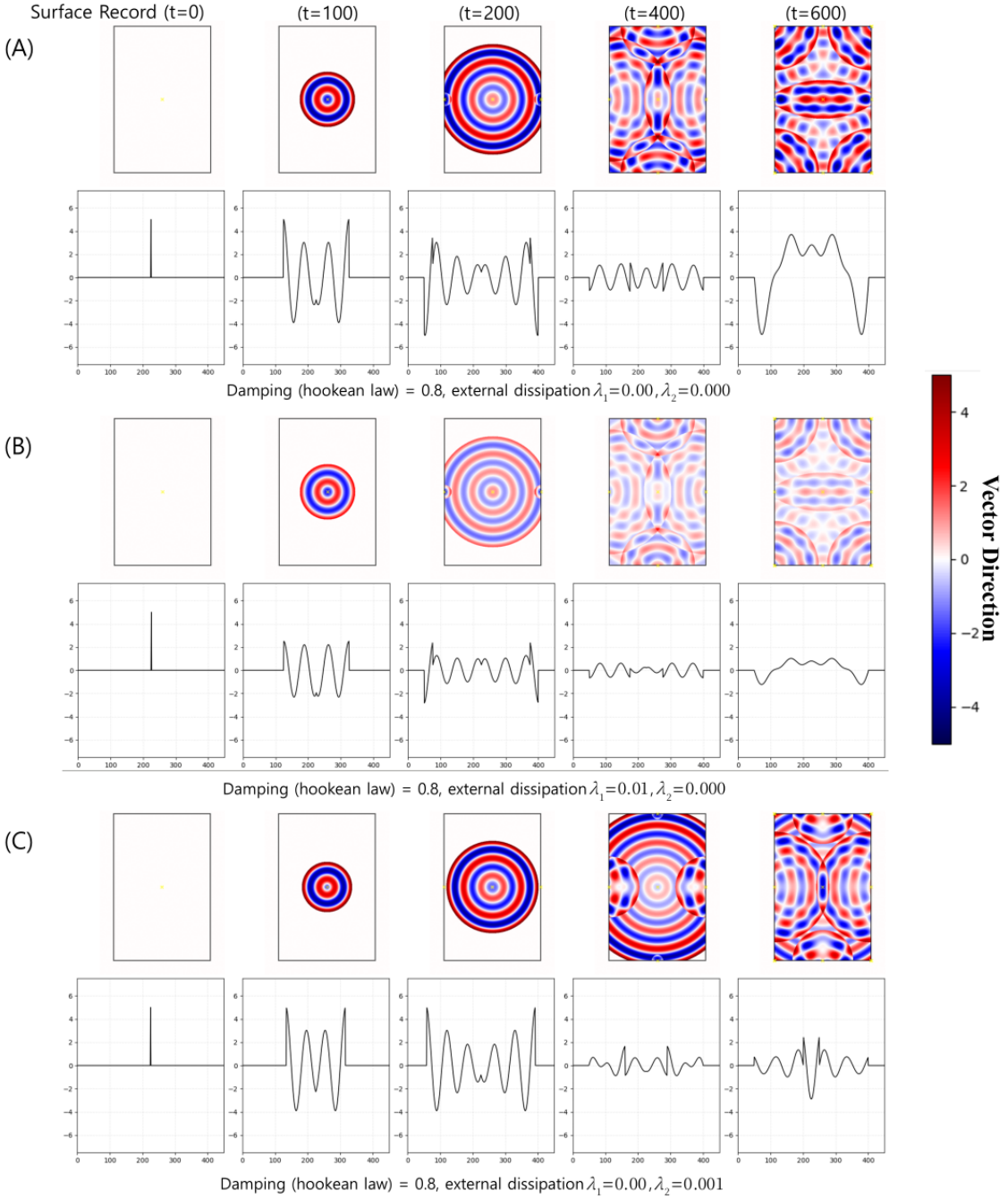


Fig. S5. Surface patterns validation of the external dissipation. (A) Baseline model without dissipation, (B) continuous spatial force decay governed by λ_1, and (C) spatiotemporal temporal delay governed by λ_2

Movie S1 (separate file). Surface patterns generated by a single causal source

Movie S2 (separate file). Surface patterns generated by a single causal source with modified parameters

Movie S3 (separate file). Surface record of a single damped source

Movie S4 (separate file). Interaction of two identical causal sources

Movie S5 (separate file). Validation of spatial propagation arrest via the beta field

Movie S6 (separate file). Numerical surface records generated by a single localized source within a restricted beta field

Movie S7A (separate file). Numerical surface records generated by a single localized source within a restricted beta field

Movie S7B (separate file). The emergent surface patterns under varying local time steps (dt = 1)

Movie S7C (separate file). The emergent surface patterns under varying local time steps (dt = 1.5)

Movie S7D (separate file). The emergent surface patterns under varying local time steps (dt = 0.7)

Movie S8 (separate file). Symmetrical geometric order through constrained causal interactions

Movie S9 (separate file). Symmetrical geometric order through constrained causal interactions (Shifts in local stiffness)

Movie S10 (separate file). Symmetrical geometric order through constrained causal interactions (Expansion of the spatial domain)

Movie S11 (separate file). Symmetrical geometric order through constrained causal interactions (Expansion of the spatial domain)

Movie S12 (separate file). Symmetrical geometric order through constrained causal interactions (Collective responses)

Movie S13 (separate file). Symmetrical geometric order through constrained causal interactions (Collective responses with increased mass)

Movie S14 (separate file). Temporal symmetry under stochastic spatial inputs

Movie S15 (separate file). Temporal symmetry under stochastic spatial inputs (Expansion of the spatial domain)

Movie S16 (separate file). Temporal symmetry under stochastic spatial and temporal inputs

Movie S17A (separate file). Surface patterns validation of the external dissipation (A)

Movie S17B (separate file). Surface patterns validation of the external dissipation (B)

Movie S17C (separate file). Surface patterns validation of the external dissipation (C)